

UI

&

MODEL

SPECIFICATION SHEET

Build It in English · Day 2

Hoja de especificación de interfaz y modelo · Día 2

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Project / Proyecto	Signal Desk
Environment / Entorno	C#-Maui
Accent colour / Color de acento	000000
Fonts / Tipografías	Cambria
Date / Fecha	2026-09-01

1. Interface controls

PANTALLA	NOMBRE DE CONTROL	TIPO	TEXTO MOSTRADO AL USUARIO	QUÉ HARÁ
Main	txtAge	TextBox	Placeholder: e.g. 25	Accepts user age input in years.
Main	txtWeight	TextBox	Placeholder: e.g. 70.5	Accepts body weight in kilograms.
Main	txtHeight	TextBox	Placeholder: e.g. 175	Accepts height in centimeters
Main	cmbActivity	ComboBox	Select activity level	Selects daily physical activity level factor.
Main	btnCalculate	Button	Calculate BMR & TDEE	Triggers input validation and executes calculation.

Main	btnClear	Button	Clear Form	Resets all input fields and resets output labels.
Main	lblBmrResult	Label	BMR: 0.00 kcal/day	Displays calculated Basal Metabolic Rate.
Main	lblTdeeResult	Label	TDEE: 0.00 kcal/day	Displays calculated Total Daily Energy Expenditure

2. Microcopy inventory

SITUACIÓN	TEXTO EN INGLÉS	COPIA EN ESPAÑOL (SOLO PARA EL INFORME)
Estado vacío	Enter your biometric data above to calculate BMR and TDEE.	Ingrese sus datos biométricos arriba para calcular BMR y TDEE.
Error de validación 1	Age must be a whole number between 15 and 100.	La edad debe ser un número entero entre 15 y 100.
Error de validación 2	Weight and height must be positive decimal numbers.	El peso y la estatura deben ser números decimales positivos.
Eliminar confirmación	Are you sure you want to clear all entered data?	¿Está seguro de que desea borrar todos los datos ingresados?
Mensaje de éxito	Calculations completed successfully.	Cálculos completados exitosamente.
Sin resultados de búsqueda	No profile matches the specified criteria.	Ningún perfil coincide con los criterios especificados.

3. Model classes

CLASE	CAMPO	TIPO	LO QUE ALMACENA	REGLA DE VALIDACIÓN
UserProfile	Age	int	User age in years	Must be integer between 15 and 100

UserProfile	WeightKg	double	Body weight in kilograms	Must be greater than 0
UserProfile	HeightCm	double	Height in centimeters	Must be greater than 0
UserProfile	Gender	string	User gender ("Male"/"Female")	Cannot be null or empty
UserProfile	ActivityLevel	string	Daily physical activity factor	Must be a valid selected item
Must be a valid selected item	BmrResult	double	Calculated basal metabolic rate	Read-only computed value

4. Methods planned

FIRMA DEL MÉTODO (INGLÉS)	QUÉ HACE	DE DONDE SE LLAMA
Method signature (English)What it doesWhere it is called frompublic double CalculateBMR(UserProfile profile)	Calculates Basal Metabolic Rate using Mifflin-St Jeor equation	Called inside btnCalculate_Click event handler
public double CalculateTDEE(double bmr, string activityLevel)	Multiplies BMR by activity factor to get total calories	Called inside btnCalculate_Click handler
public bool ValidateInputs(out string errorMessage)	Verifies that all fields contain valid numerical values	Called before executing calculation logic
public void ResetFields()	Clears controls and resets values to default	Called inside btnClear_Click event handler

5. Problems and solutions

Handling user input parsing errors when textboxes were left empty or non-numeric values were typed. Solved by implementing `double.TryParse` validation methods with specific localized English error messages before passing data to the model.

Manejo de errores de conversión al ingresar valores no numéricos o dejar campos vacíos. Se resolvió implementando validaciones con `double.TryParse` y mostrando mensajes de error claros en inglés antes de enviar los datos al modelo.

Evidence: attach screenshot EV-02 of the finished interface and paste one model class with its comments.

Adjunte la captura EV-02 de la interfaz terminada y pegue una clase del modelo con sus comentarios.